

Q1 - 25 July - Shift 1

For any real number x , let $[x]$ denote the largest integer less than or equal to x . Let f be a real valued function defined on the interval

$$[-10, 10] \text{ by } f(x) = \begin{cases} x - [x], & \text{if } (x) \text{ is odd} \\ 1 + [x] - x & \text{if } (x) \text{ is even} \end{cases}$$

Then the value of $\frac{\pi^2}{10} \int_{-10}^{10} f(x) \cos \pi x dx$ is :

- (A) 4 (B) 2
(C) 1 (D) 0

Space for your notes:

Q2 - 25 July - Shift 2

$$\lim_{n \rightarrow \infty} \frac{1}{2^n} \left(\frac{1}{\sqrt{1 - \frac{1}{2^n}}} + \frac{1}{\sqrt{1 - \frac{2}{2^n}}} + \frac{1}{\sqrt{1 - \frac{3}{2^n}}} + \dots + \frac{1}{\sqrt{1 - \frac{2^n - 1}{2^n}}} \right)$$

is equal to

- (A) $\frac{1}{2}$ (B) 1
(C) 2 (D) -2

Space for your notes:

Q3 - 25 July - Shift 2

Questions

MathonGo

Let $[t]$ denote the greatest integer less than or equal to t . Then the value of the integral

$\int_{-3}^{101} ([\sin(\pi x)] + e^{\cos(2\pi x)}) dx$ is equal to

- (A) $\frac{52(1-e)}{e}$ (B) $\frac{52}{e}$
 (C) $\frac{52(2+e)}{e}$ (D) $\frac{104}{e}$

Space for your notes:

Q4 - 25 July - Shift 2

Let f be a twice differentiable function on $\mathbb{R}$.

If $f'(0) = 4$ and

$$f(x) + \int_0^x (x-t)f'(t)dt = (e^{2x} + e^{-2x})\cos 2x + \frac{2}{a}x,$$

then $(2a+1)^5 \cdot a^2$ is equal to _____

Space for your notes:

Q5 - 25 July - Shift 2

Let $a_n = \int_{-1}^n \left(1 + \frac{x}{2} + \frac{x^2}{2} + \frac{x^3}{3} + \dots + \frac{x^{n-1}}{n} \right) dx$

for $n \in \mathbb{N}$. Then the sum of all the elements of the set $\{n \in \mathbb{N} : a_n \in (2, 30)\}$ is _____

Space for your notes:

Q6 - 26 July - Shift 1

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Questions

MathonGo

If $a = \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{2n}{n^2 + k^2}$ and $f(x) =$

Space for your notes:

$\sqrt{\frac{1 - \cos x}{1 + \cos x}}$, $x \in (0, 1)$, then :

(A) $2\sqrt{2}f\left(\frac{a}{2}\right) = f'\left(\frac{a}{2}\right)$

(B) $f\left(\frac{a}{2}\right)f'\left(\frac{a}{2}\right) = \sqrt{2}$

(C) $\sqrt{2}f\left(\frac{a}{2}\right) = f'\left(\frac{a}{2}\right)$

(D) $f\left(\frac{a}{2}\right) = \sqrt{2}f'\left(\frac{a}{2}\right)$

Q7 - 26 July - Shift 1

If $n(2n+1) \int_0^1 (1-x^n)^{2n} dx = 1177 \int_0^1 (1-x^n)^{2n+1} dx$, then

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$n \in \mathbb{N}$ is equal to _____

Q8 - 26 July - Shift 2

$\int_0^{20\pi} (|\sin x| + |\cos x|)^2 dx$ is equal to :-

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(A) $10(\pi + 4)$ (B) $10(\pi + 2)$

(C) $20(\pi - 2)$ (D) $20(\pi + 2)$

Q9 - 27 July - Shift 1

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Questions

MathonGo

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined as

Space for your notes:

$$f(x) = a \sin\left(\frac{\pi[x]}{2}\right) + [2-x], \quad a \in \mathbb{R}, \text{ where } [t]$$

is the greatest integer less than or equal to t . If

$$\lim_{x \rightarrow -1} f(x) \text{ exists, then the value of } \int_0^4 f(x) dx \text{ is}$$

equal to :

(A) -1 (B) -2

(C) 1 (D) 2

Q10 - 27 July - Shift 2

Let $f(x) = 2 + |x| - |x-1| + |x+1|$, $x \in \mathbb{R}$.

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Consider

$$(S1): f'\left(-\frac{3}{2}\right) + f'\left(-\frac{1}{2}\right) + f'\left(\frac{1}{2}\right) + f'\left(\frac{3}{2}\right) = 2$$

$$(S2): \int_{-2}^2 f(x) dx = 12$$

Then,

(A) both (S1) and (S2) are correct

(B) both (S1) and (S2) are wrong

(C) only (S1) is correct

(D) only (S2) is correct

Q11 - 27 July - Shift 2

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Questions

MathonGo

$$\int_0^2 \left(|2x^2 - 3x| + \left[x - \frac{1}{2} \right] \right) dx,$$

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where $[t]$ is the greatest integer function, is equal to:

- (A) $\frac{7}{6}$ (B) $\frac{19}{12}$ (C) $\frac{31}{12}$ (D) $\frac{3}{2}$

Q12 - 27 July - Shift 2

Let $f(x) = \min \{[x - 1], [x - 2], \dots, [x - 10]\}$ where $[t]$ denotes the greatest integer $\leq t$. Then

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$$\int_0^{10} f(x) dx + \int_0^{10} (f(x))^2 dx + \int_0^{10} |f(x)| dx \text{ is equal to } \underline{\hspace{2cm}}.$$

Q13 - 27 July - Shift 2

Let f be a differentiable function satisfying

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$$f(x) = \frac{2}{\sqrt{3}} \int_0^{\sqrt{3}} f\left(\frac{\lambda^2 x}{3}\right) d\lambda, x > 0 \text{ and } f(1) = \sqrt{3}. \text{ If}$$

$y=f(x)$ passes through the point $(\alpha, 6)$, then α is equal to $\underline{\hspace{2cm}}$.

Q14 - 28 July - Shift 1

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Questions

MathonGo

The minimum value of the twice differentiable function $f(x) = \int_0^x e^{x-t} f'(t) dt - (x^2 - x + 1)e^x$, $x \in \mathbb{R}$,

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is :

(A) $-\frac{2}{\sqrt{e}}$ (B) $-2\sqrt{e}$

(C) $-\sqrt{e}$ (D) $\frac{2}{\sqrt{e}}$

Q15 - 28 July - Shift 1

If $\int_0^{\sqrt{3}} \frac{15x^3}{\sqrt{1+x^2} + \sqrt{(1+x^2)^3}} dx = \alpha\sqrt{2} + \beta\sqrt{3}$, where

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α, β are integers, then $\alpha + \beta$ is equal to

Q16 - 28 July - Shift 2

Let $I_n(x) = \int_0^x \frac{1}{(t^2 + 5)^n} dt$, $n = 1, 2, 3, \dots$. Then

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(A) $50I_6 - 9I_5 = xI_5'$ (B) $50I_6 - 11I_5 = xI_5'$
(C) $50I_6 - 9I_5 = I_5'$ (D) $50I_6 - 11I_5 = I_5'$

Q17 - 28 July - Shift 2

The value of the integral $\int_0^{\frac{\pi}{2}} 60 \frac{\sin(6x)}{\sin x} dx$ is equal

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to

Q18 - 29 July - Shift 1

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The integral $\int_0^{\frac{\pi}{2}} \frac{1}{3+2\sin x + \cos x} dx$ is equal to:

- (A) $\tan^{-1}(2)$ (B) $\tan^{-1}(2) - \frac{\pi}{4}$
(C) $\frac{1}{2} \tan^{-1}(2) - \frac{\pi}{8}$ (D) $\frac{1}{2}$

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Q19 - 29 July - Shift 1

If $f(\alpha) = \int_1^{\alpha} \frac{\log_{10} t}{1+t} dt, \alpha > 0$, then $f(e^3) + f(e^{-3})$ is equal to :

- (A) 9 (B) $\frac{9}{2}$
(C) $\frac{9}{\log_e(10)}$ (D) $\frac{9}{2 \log_e(10)}$

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Q20 - 29 July - Shift 2

If $[t]$ denotes the greatest integer $\leq t$, then the value of $\int_0^1 [2x - |3x^2 - 5x + 2| + 1] dx$ is:

- (A) $\frac{\sqrt{37} + \sqrt{13} - 4}{6}$ (B) $\frac{\sqrt{37} - \sqrt{13} - 4}{6}$
(C) $\frac{-\sqrt{37} - \sqrt{13} + 4}{6}$ (D) $\frac{-\sqrt{37} + \sqrt{13} + 4}{6}$

Space for your notes:

Answer Key

Q1 (A)

Q2 (C)

Q3 (B)

Q4 (8)

Q5 (5)

Q6 (C)

Q7 (24)

Q8 (D)

Q9 (B)

Q10 (D)

Q11 (B)

Q12 (385)

Q13 (12)

Q14 (A)

Q15 (10)

Q16 (A)

Q17 (104)

Q18 (B)

Q19 (D)

Q20 (A)

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Q1 (A)

$f(x)$ is periodic function whose period is 2

$$\begin{aligned} \frac{\pi^2}{10} \int_{-10}^{10} f(x) \cos \pi x dx &= \frac{\pi^2}{10} \times 10 \int_0^2 f(x) \cos \pi x dx \\ &= \pi^2 \left(\int_0^1 (1-x) \cos \pi x dx + \int_1^2 (x-1) \cos \pi x dx \right) \end{aligned}$$

Using by parts

$$= \pi^2 \times \frac{4}{\pi^2} = 4$$

Q2 (C)

(C)

Q3 (B)

(B)

Q4 (8)

(8)

Q5 (5)

(5)

Q6 (C)

$$a = \frac{1}{n} \sum_{k=1}^n \frac{2}{1 + \left(\frac{k}{n}\right)^2} = \int_0^1 \frac{2}{1 + x^2} dx = \frac{\pi}{2}$$

$$f(x) = \tan\left(\frac{x}{2}\right); x \in (0, 1)$$

$$f\left(\frac{\pi}{4}\right) = \sqrt{2} - 1$$

$$f'\left(\frac{\pi}{4}\right) = \frac{1}{2} \sec^2\left(\frac{\pi}{8}\right) = \frac{\sqrt{2}}{\sqrt{2} + 1}$$

$$f'\left(\frac{\pi}{4}\right) = \sqrt{2} f\left(\frac{\pi}{4}\right)$$

Q7 (24)

$$\text{Let } I_1 = \int_0^1 (1 - x^n)^{2n} dx, I_2 = \int_0^1 (1 - x^n)^{2n+1} dx$$

$$I_2 = \int_0^1 (1 - x^n)^{2n+1} \cdot 1 dx$$

$$= (1 - x^n)^{2n+1} \cdot x \Big|_0^1 - \int_0^1 (2n+1)(1 - x^n)^{2n} (-nx^{n-1}) x dx$$

$$I_2 = -n(2n+1) \{I_2 - I_1\}$$

$$(2n^2 + n + 1)I_2 = n(2n+1)I_1$$

$$\frac{I_1}{I_2} = \frac{2n^2 + n + 1}{n(2n+1)} = \frac{1177}{n(2n+1)}$$

$$\Rightarrow 2n^2 + n - 1176 = 0 \Rightarrow n = 24$$

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Q8 (D)

$$I = \int_0^{20\pi} (|\sin x| + |\cos x|)^2 dx \quad ; \text{ (Jack property)}$$

$$I = 40 \int_0^{\pi/2} (\sin x + \cos x)^2 dx$$

$$I = 40 \int_0^{\pi/2} (1 + \sin 2x) dx$$

$$I = 20[\pi + 2]$$

Q9 (B)

$$\lim_{x \rightarrow -1^+} a \sin\left(\pi \frac{[x]}{2}\right) + [2 - x] = -a + 2$$

$$\lim_{x \rightarrow -1^-} a \sin\left(\pi \frac{[x]}{2}\right) + [2 - x] = 0 + 3 = 3$$

$$\lim_{x \rightarrow -1} f(x) \text{ exist when } a = -1$$

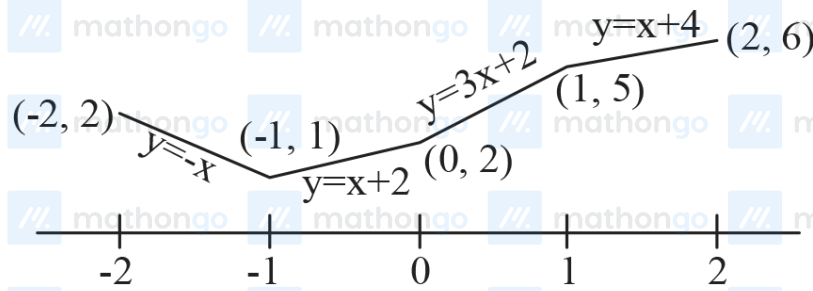
Now,

$$\int_0^4 f(x) dx = \int_0^1 f(x) dx + \int_1^2 f(x) dx + \int_2^3 f(x) dx + \int_3^4 f(x) dx$$

$$= \int_0^1 (0 + 1) dx + \int_1^2 (-1 + 0) dx + \int_2^3 (0 - 1) dx + \int_3^4 (1 - 2) dx$$

$$= 1 - 1 - 1 - 1 = -2$$

Q10 (D)



$$(S1): f'\left(-\frac{3}{2}\right) + f'\left(-\frac{1}{2}\right) + f'\left(\frac{1}{2}\right) + f'\left(\frac{3}{2}\right) = 4$$

$$(S2): \int_{-2}^2 f(x) dx = 12$$

$\therefore$ (D)

Q11 (B)

$$\int_0^2 |2x^2 - 3x| dx$$

$$= \int_0^{\frac{3}{2}} (3x - 2x^2) dx + \int_{\frac{3}{2}}^2 (2x^2 - 3x) dx = \frac{19}{12}.$$

$$\int_0^2 \left[x - \frac{1}{2} \right] dx = \int_{-\frac{1}{2}}^{\frac{3}{2}} [t] dt$$

$$= \int_{-\frac{1}{2}}^0 (-1) dt + \int_0^1 0 \cdot dt + \int_1^{\frac{3}{2}} 1 \cdot dt = 0.$$

Q12 (385)

$$f(x) = [x] - 10$$

$$\int_0^{10} f(x) \cdot dx = -10 - 9 - 8 - \dots - 1$$

$$= -\frac{10 \cdot 11}{2} = -55$$

$$\int_0^{10} (f(x))^2 dx = 10^2 + 9^2 + 8^2 + \dots + 1^2$$

$$= \frac{10 \cdot 11 \cdot 21}{6} = 385$$

$$\int_0^{10} |f(x)| = 10 + 9 + 8 + \dots + 1$$

$$= \frac{10 \cdot 11}{2} = 55$$

$$= -55 + 385 + 55 = 385$$

Q13 (12)

$$\text{Let, } \frac{\lambda^2 x}{3} = t$$

$$\Rightarrow \frac{2\lambda x}{3} d\lambda = dt$$

$$\Rightarrow d\lambda = \frac{3}{2} \cdot \frac{1}{\sqrt{x}} \frac{1}{\sqrt{3\lambda}} dt$$

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$$\Rightarrow d\lambda = \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{x}} \cdot \frac{dt}{\sqrt{t}}$$

$$\text{So, } f(x) = \frac{1}{\sqrt{x}} \int_0^x \frac{f(t)}{\sqrt{t}} dt$$

$$\Rightarrow \sqrt{x} \cdot f'(x) + \frac{f(x)}{2\sqrt{x}} = \frac{f(x)}{\sqrt{x}}$$

$$\Rightarrow \sqrt{x} \cdot f'(x) = \frac{f(x)}{2\sqrt{x}}$$

$$\Rightarrow \frac{dy}{y} = \frac{dx}{2x}$$

$$\Rightarrow \ln y = \frac{1}{2} \ln x + c \Rightarrow f(x) = \sqrt{x}$$

$$\Rightarrow y = \sqrt{3x} \quad \{\text{as } f(1) = \sqrt{3}\}$$

$$\text{So, } f(x) = \sqrt{3x}$$

$$\text{Now, } f(\alpha) = 6 \Rightarrow 36 = 3\alpha$$

$$\Rightarrow \alpha = 12$$

Q14 (A)

$$f(x) = e^x \cdot \int_0^x \frac{f'(t)}{e^t} dt$$

$$f'(x) = e^x \cdot \int_0^x \frac{f'(t)}{e^t} dt + e^x \cdot \frac{f'(x)}{e^x}$$

$$-\left[(2x-1) \cdot e^x + (x^2 - x + 1) \cdot e^x\right]$$

$$\int_0^x \frac{f'(t)}{e^t} dt = x^2 + x$$

$$\frac{f'(x)}{e^x} = 2x + 1$$

$$f'(x) = (2x + 1) \cdot e^x$$

$$f'(x) = 0 \Rightarrow x = -\frac{1}{2}$$

$$f(x) = (2x + 1) \cdot e^x - 2e^x + C$$

$$\left| \begin{array}{l} f(0) = -1 \end{array} \right.$$

$$-1 = 1 - 2 + C$$

$$C = 0$$

$$f(x) = e^x (2x - 1)$$

$$f\left(-\frac{1}{2}\right) = \frac{-2}{\sqrt{e}}$$

Q15 (10)

$$\text{Put } 1 + x^2 = t^2$$

$$2x dx = 2t dt$$

$$X dx = t dt$$

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$$15 \int_1^2 \frac{t(t^2 - 1)}{t\sqrt{1+t}} dt$$

Put $1 + t = u^2$

$$dt = 2u du$$

$$15 \int_{\sqrt{2}}^{\sqrt{3}} \frac{(u^2 - 1)^2 - 1}{u} \times 2u du$$

$$30 \int_{\sqrt{2}}^{\sqrt{3}} (u^4 - 2u^2) du$$

$$30 \left(\frac{u^5}{5} - \frac{2u^3}{3} \right)_{\sqrt{2}}^{\sqrt{3}}$$

$$30 \left[\frac{1}{5} (\sqrt{3}^5 - \sqrt{2}^5) - \frac{2}{3} (\sqrt{3}^3 - \sqrt{2}^3) \right]$$

$$30 \left[\frac{1}{5} (9\sqrt{3} - 4\sqrt{2}) - \frac{2}{3} (3\sqrt{3} - 2\sqrt{2}) \right]$$

$$30 \left[-\frac{1}{5} \times \sqrt{3} + \frac{8}{15} \sqrt{2} \right]$$

$$-6\sqrt{3} + 16\sqrt{2} = \alpha\sqrt{2} + \beta\sqrt{3}$$

$$\alpha = 16, \beta = -6$$

$$\therefore \alpha + \beta = 10$$

Q16 (A)

$$I_n(x) = \int_0^x \frac{dt}{(t^2 + 5)^n}$$

Applying integral by parts

$$I_n(x) = \left[\frac{t}{(t^2 + 5)^n} \right]_0^x - \int_0^x n(t^2 + 5)^{-n-1} \cdot 2t^2$$

$$I_n(x) = \frac{x}{(x^2 + 5)^n} + 2n \int_0^x \frac{t^2}{(t^2 + 5)^{n+1}} dt$$

$$I_n(x) = \frac{x}{(x^2 + 5)^n} + 2n \int_0^x \frac{(t^2 + 5) - 5}{(t^2 + 5)^{n+1}} dt$$

$$I_n(x) = \frac{x}{(x^2 + 5)^n} + 2n I_n(x) - 10n I_{n+1}(x)$$

$$10n I_{n+1}(x) + (1 - 2n)I_n(x) = \frac{x}{(x^2 + 5)^n}$$

Put $n = 5$

Q17 (104)

$$I = 60 \int_0^{\pi/2} \left(\frac{\sin 6x - \sin 4x}{\sin x} + \frac{\sin 4x - \sin 2x}{\sin x} + \frac{\sin 2x}{\sin x} \right) dx$$

$$I = 60 \int_0^{\pi/2} (2 \cos 5x + 2 \cos 3x + 2 \cos x) dx$$

$$I = 60 \left(\frac{2}{5} \sin 5x + \frac{2}{3} \sin 3x + 2 \sin x \right) \Big|_0^{\pi/2} = 104$$

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Q18 (B)

$$I = \int_0^{\frac{\pi}{2}} \frac{dx}{3 + 2 \sin x + \cos x} = \int_0^{\frac{\pi}{2}} \frac{\sec^2 \frac{x}{2} \cdot dx}{2 \tan^2 \frac{x}{2} + 4 \tan \frac{x}{2} + 4}$$

Put $\tan \frac{x}{2} = t$, so

$$I = \int_0^1 \frac{dt}{(t+1)^2 + 1} = \tan^{-1}(t+1) \Big|_0^1 = \tan^{-1} 2 - \frac{\pi}{4}$$

Q19 (D)

$$f(e^3) = \int_1^{e^3} \frac{\ln t}{\ln 10(1+t)} dt \dots (1)$$

$$f(\alpha) = \int_1^{\alpha} \frac{\ln t}{(\ln 10)(1+t)} dt$$

$$t = \frac{1}{x} \Rightarrow x = \frac{1}{t}$$

$$dt = \frac{-1}{x^2} dx$$

$$= \int_1^{\alpha} \frac{-\ln x}{(\ln 10) \left(1 + \frac{1}{x}\right)} \left(-\frac{1}{x^2}\right) dx$$

$$f(\alpha) = \frac{1}{\ln 10} \int_1^{\alpha} \frac{\ln x}{x(x+1)} dx$$

$$f(e^{-3}) = \frac{1}{\ln 10} \int_1^{e^3} \frac{\ln t}{1+t} dt \dots (2)$$

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Ans and Solutions

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$$f(e^3) + f(e^{-3})$$

$$= \left(\frac{1}{\ln 10} \right) \int_1^{e^3} \frac{\ln t}{(1+t)} \left[1 + \frac{1}{t} \right] dt$$

$$= \left(\frac{1}{\ln 10} \right) \int_1^3 \frac{\ln t}{t} dt$$

$$\ln t = r$$

$$\frac{dt}{t} = dr$$

$$= \frac{1}{\ln 10} \int_0^3 r dr$$

$$= \left(\frac{1}{\ln 10} \right) \left(\frac{r^2}{2} \right) \Big|_0^3$$

$$= \left(\frac{1}{\log 10} \right) \left(\frac{9}{2} \right)$$

$$= \frac{9}{2 \log_e 10}$$

Q20 (A)

$$I = \int_0^1 [2x - |3x^2 - 3x - 2x + 2| + 1] dx$$

$$I = \int_0^1 [2x - |(3x - 2)(x + 1)|] dx + \int_0^1 1 dx$$

$$I = \int_0^{2/3} [(2x - (3x^2 - 5x + 2))] dx + \int_{2/3}^1 (2x + (3x^2 - 5x + 2)) dx + 1$$

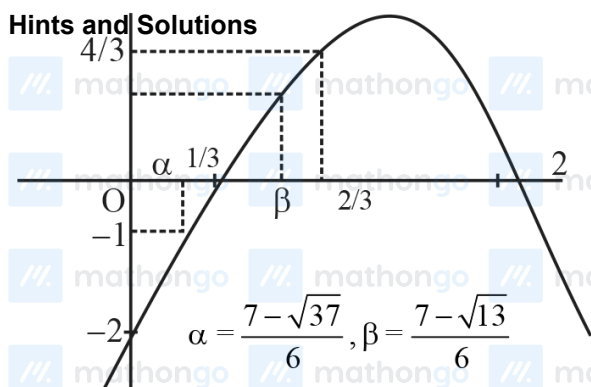
$$I = \int_0^{2/3} [-3x^2 + 7x - 2] dx + \int_{2/3}^1 (3x^2 - 3x + 2) dx + 1$$

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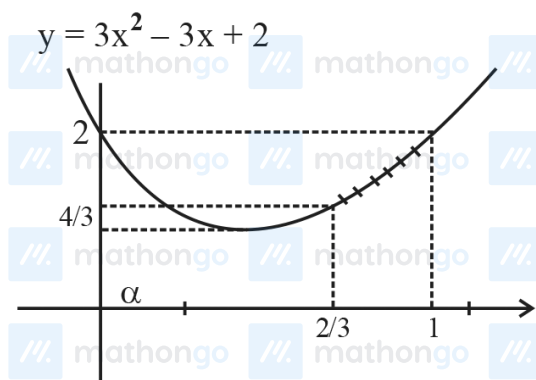
Hints and Solutions

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$$\int_0^{\alpha} (-2)dx + \int_{\alpha}^{1/3} (-1)dx + \int_{1/3}^{\beta} 0dx + \int_{\beta}^{2/3} 1dx$$

$$= -2\alpha - \left(\frac{1}{3} - \alpha\right) + \frac{2}{3} - \beta = -\alpha - \beta + \frac{1}{3}$$



When $x \in \left(\frac{2}{3}, 1\right)$

$$3x^2 - 3x + 2 \in \left(\frac{4}{3}, 2\right)$$

$$[3x^2 - 3x + 2] = 1$$

$$\therefore \int_{2/3}^1 [3x^2 - 3x + 2]dx = 1 \left(1 - \frac{2}{3}\right) = \frac{1}{3}$$

$$\text{Hence } I = \left(\frac{1}{3} - (\alpha + \beta)\right) + \left(\frac{1}{3}\right) + 1$$

$$= \frac{5}{3} - \left(\frac{7 - \sqrt{37}}{6} + \frac{7 - \sqrt{13}}{6}\right)$$

$$= \frac{-2}{3} + \frac{\sqrt{37} + \sqrt{13}}{6}$$

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